The documentation and process conversion measures necessary to comply with this revision shall be completed by 3 September 2002 .

INCH-POUND

MIL-PRF-19500/251K
3 June 2002
SUPERSEDING
MIL-PRF-19500/251J
10 August 1998

PERFORMANCE SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, SWITCHING
TYPES 2N2218, 2N2218A, 2N2218AL, 2N2219, 2N2219A, AND 2N2219AL
JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for NPN, silicon, switching transistors. Four levels of product assurance are provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1 (similar to TO-39, TO-5).

* 1.3 Maximum ratings.

Types	P_T T_A = +25°C (1)	P_T T_C = +25°C (2)	V_{CBO}	V_{CEO}	V_{EBO}	I_C	T_{STG} and T_{OP}	$R_{\theta JC}$ max
	W	W	V dc	V dc	V dc	mA dc		°C/W
2N2218, 2N2219	0.8	3.0	60	30	5	800	-55° to +200°C	59
2N2218A, 2N2219A	0.8	3.0	75	50	6	800	-55° to +200°C	59
2N2218AL, 2N2219AL	0.8	3.0	75	50	6	800	-55° to +200°C	59

(1) Derate linearly 4.6 mW/°C for T_A > +25°C.
(2) Derate linearly 17.0 mW/°C for T_C > +25°C.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Supply Center Columbus, ATTN: DSCC-VAC, P. O. Box 3990, Columbus, OH 43216-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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* 1.4 Primary electrical characteristics.

Types	h_{FE} at V_{CE} = 10 V dc									
	h_{FE1} I_C = 100 µA dc		h_{FE2} I_C = 1.0 mA dc		h_{FE3} I_C = 10 mA dc		h_{FE4} (1) I_C = 150 mA dc		h_{FE5} (1) I_C = 500 mA dc	
	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
2N2218	20		25	150	35		40	120	20	
2N2219	35		50	325	75		100	300	30	
2N2218A	30		35	150	40		40	120	20	
2N2219A	50		75	325	100		100	300	30	
2N2218AL	30		35	150	40		40	120	20	
2N2219AL	50		75	325	100		100	300	30	

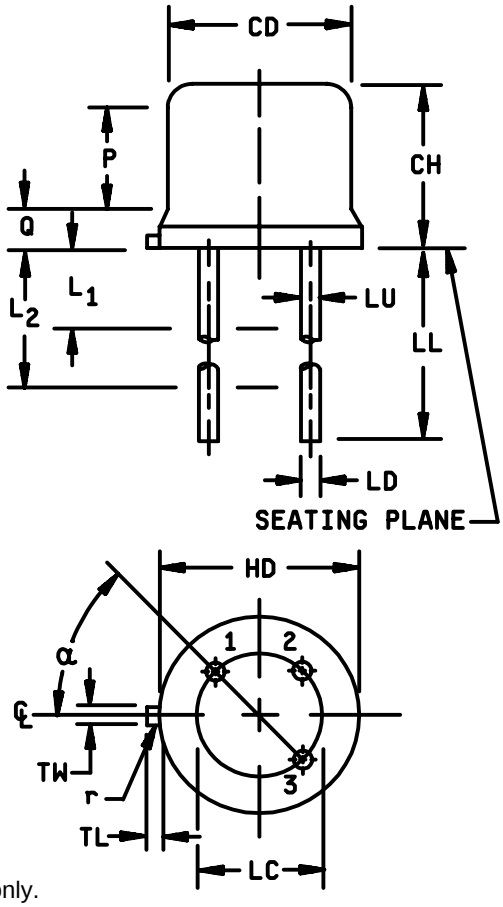
Types	| h_{fe} | I_C = 20 mA dc V_{CE} = 20 V dc f = 100 MHz		C_{obo} I_E = 0, V_{CE} = 10 V dc 100 kHz ≤ f ≤ 1 MHz		Switching t_{on} See figure 2		Switching t_{off} See figure 3	
			pF		ns		ns	
	Min	Max	Min	Max	Min	Max	Min	Max
2N2218	2.5	12.0		8		40		250
2N2219	2.5	12.0		8		40		250
2N2218A	2.5	12.0		8		35		300
2N2219A	2.5	12.0		8		35		300
2N2218AL	2.5	12.0		8		35		300
2N2219AL	2.5	12.0		8		35		300

Types	$V_{CE(sat)1}$ (1) I_C = 150 mA dc I_B = 15 mA dc		$V_{CE(sat)2}$ (1) I_C = 500 mA dc I_B = 50 mA dc		$V_{BE(sat)1}$ (1) I_C = 150 mA dc I_B = 15 mA dc		$V_{BE(sat)2}$ (1) I_C = 500 mA dc I_B = 50 mA dc	
	V dc		V dc		V dc		V dc	
	min	max	min	max	min	max	min	max
2N2218		0.4		1.6	0.6	1.3		2.6
2N2219		0.4		1.6	0.6	1.3		2.6
2N2218A		0.3		1.0	0.6	1.2		2.0
2N2219A		0.3		1.0	0.6	1.2		2.0
2N2219AL		0.3		1.0	0.6	1.2		2.0
2N2219AL		0.3		1.0	0.6	1.2		2.0

(1) Pulsed (see 4.5.1).

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Symbol	Dimensions				Note
	Inches		Millimeters		
	Min	Max	Min	Max	
CD	.305	.335	7.75	8.51	
CH	.240	.260	6.12	6.60	
HD	.335	.370	8.51	9.40	
LC	.200 TP		5.08 TP		7
LD	.016	.019	0.41	0.48	8,9
LL	See note 14				
LU	.016	.019	0.41	0.48	8,9
L1		.050		1.27	8,9
L2	.250		6.35		8,9
P	.100		2.54		7
Q		.030		0.76	5
TL	.029	.045	0.74	1.14	3,4
TW	.028	.034	0.71	0.86	3
r		.010		0.25	10
α	45° TP		45° TP		7



NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Beyond r (radius) maximum, TW shall be held for a minimum length of .011 (0.28 mm).
4. Dimension TL measured from maximum HD.
5. Body contour optional within zone defined by HD, CD, and Q.
6. CD shall not vary more than .010 inch (0.25 mm) in zone P. This zone is controlled for automatic handling.
7. Leads at gauge plane .054 +.001 -.000 inch (1.37 +0.03 -0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC.
8. Dimension LU applies between L_1 and L_2. Dimension LD applies between L_2 and LL minimum. Diameter is uncontrolled in L_1 and beyond LL minimum.
9. All three leads.
10. The collector shall be internally connected to the case.
11. Dimension r (radius) applies to both inside corners of tab.
12. In accordance with ANSI Y14.5M, diameters are equivalent to ϕx symbology.
13. Lead 1 = emitter, lead 2 = base, lead 3 = collector.
14. For L suffix devices (T0-5), dimension LL = 1.5 inches (38.10 mm) min. and 1.75 inches (44.45 mm) max. For non-L suffix types (T0-39), dimension LL = .5 inches (12.70 mm) min. and .750 inches (19.05 mm) max.

FIGURE 1. Physical dimensions (similar to TO-39, T0-5).

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

DEPARTMENT OF DEFENSE

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Document Automation and Production Services (DAPS), Building 4D (DPM-DODSSP), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 General. The requirements for acquiring the product described herein shall consist of this document and MIL-PRF-19500.

* 3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

* 3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

* 3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figure 1.

* 3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

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* 3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

* 3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

* 3.7 Electrical test requirements. The electrical test requirements shall be group A as specified herein.

* 3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

* 4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with table IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
3c	Thermal impedance (see 4.3.2).	Thermal impedance (see 4.3.2).
9	I_{CBO2}, h_{FE2}	Not applicable
10	48 hours minimum	48 hours minimum
11	I_{CBO2}; h_{FE2}; ΔI_{CBO2} = 100 percent of initial value or 5 nA dc, whichever is greater. Δh_{FE2} = ±15 percent	I_{CBO2}, h_{FE4}
12	See 4.3.1 240 hours minimum	See 4.3.1 80 hours minimum
13	Subgroups 2 and 3 of table I herein; ΔI_{CBO2} = 100 percent of initial value or 5 nA dc, whichever is greater; Δh_{FE2} = ±15 percent	Subgroup 2 of table I herein; ΔI_{CBO2} = 100 percent of initial value or5 nA dc, whichever is greater; Δh_{FE4} = ±15 percent

* 4.3.1 Power burn-in test conditions. Power burn-in conditions are as follows: T_A = room ambient as defined in the general requirements of 4.5 of MIL-STD-750. V_{CB} = 10 - 30 V dc; adjust the power to achieve T_J = +135°C minimum. The minimum power dissipated shall be 75 percent of the rated power. NOTE: No heat sink or forced air cooling on the devices shall be permitted.

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4.3.2 Thermal impedance ($Z_{\theta JX}$ measurements). The $Z_{\theta JX}$ measurements shall be performed in accordance with MIL-STD-750, Method 3131 and as follows:

a. I_M measurement current----------5 mA.

b. I_H forward heating current ----------200 mA (min).

c. t_H heating time ----------25 - 30 ms.

d. t_{md} measurement delay time -------60 µs max.

e. V_{CE} collector-emitter voltage -------10 V dc minimum.

The maximum limit for $Z_{\theta JX}$ under these test conditions are $Z_{\theta JX}$ (max) = 72°C/W.

* 4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein. If alternate screening is being performed in accordance with MIL-PRF-19500, a sample of screened devices shall be submitted to and pass the requirements of group A1 and A2 inspection only (table VIb, group B, subgroup 1 is not required to be performed again if group B has already been satisfied in accordance with 4.4.2).

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VIa (JANS) of MIL-PRF-19500 and 4.4.2.1. Electrical measurements (end-points) and delta requirements shall be in accordance with group A, subgroup 2 and 4.5.3 herein. See 4.4.2.2 for JAN, JANTX, and JANTXV group B testing. Electrical measurements (end-points) and delta requirements for JAN, JANTX, and JANTXV shall be after each step in 4.4.2.2 and shall be in accordance with group A, subgroup 2 and 4.5.3 herein.

* 4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
B4	1037	V_{CB} = 10 V dc; No heat sink or forced-air cooling on devices shall be permitted.
B5	1027	V_{CB} = 10 - 30 V dc; $P_D \geq$ 100 percent of maximum rated P_T (see 1.3). (NOTE: If a failure occurs, resubmission shall be at the test conditions of the original sample.) Option 1: 96 hours minimum sample size in accordance with table VIa of MIL-PRF-19500, adjust T_A or P_D to achieve T_J = +275°C minimum. Option 2: 216 hours minimum, sample size = 45, c = 0; adjusted T_A or P_D to achieve a T_J = +225°C minimum.

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* 4.4.2.2 Group B inspection, (JAN, JANTX, and JANTXV). Separate samples may be used for each step. In the event of a group B failure, the manufacturer may pull a new sample at double size from either the failed assembly lot or from another assembly lot from the same wafer lot. If the new “assembly lot” option is exercised, the failed assembly lot shall be scrapped.

Step	Method	Condition
1	1039	Steady-state life: Test condition B, 340 hours, V_{CB} = 10 - 30 V dc, T_J = +150°C min. External heating of the device under test to achieve T_J = +150°C minimum is allowed provided that a minimum of 75 percent of rated power is dissipated. No heat sink or forced-air cooling on the devices shall be permitted. n = 45 devices, c = 0
2	1039	The steady-state life test of step 1 shall be extended to 1,000 hrs for each die design. Samples shall be selected from a wafer lot every twelve months of wafer production. Group B, step 2 shall not be required more than once for any single wafer lot. n = 45, c = 0.
3	1032	High-temperature life (non-operating), T_A = +200°C, t = 340 hours. n = 22, c = 0

4.4.2.3 Group B sample selection. Samples selected from group B inspection shall meet all of the following requirements:

a. For JAN, JANTX, and JANTXV samples shall be selected randomly from a minimum of three wafers (or from each wafer in the lot) from each wafer lot. For JANS, samples shall be selected from each inspection lot. See MIL-PRF-19500.

b. Must be chosen from an inspection lot that has been submitted to and passed group A, subgroup 2, conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for life test (subgroups B4 and B5 for JANS, and group B for JAN, JANTX, and JANTXV) may be pulled prior to the application of final lead finish.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500, and in 4.4.3.1 (JANS) and 4.4.3.2 (JAN, JANTX, and JANTXV) herein for group C testing. Electrical measurements (end-points) and delta requirements shall be in accordance with group A, subgroup 2 and 4.5.3 herein.

* 4.4.3.1 Group C inspection, table VII (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
C2	2036	Test condition E.
C5	3131	See 4.5.2 ($R_{\theta JC}$).
C6	1026	1,000 hours at V_{CB} = 10 V dc; T_J = +150°C min. External heating of the device under test to achieve T_J = +150°C minimum is allowed provided that a minimum of 75 percent of rated power is dissipated. No heat sink or forced-air cooling on device shall be permitted.

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* 4.4.3.2 Group C inspection, table VII (JAN, JANTX, and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Condition
C2	2036	Test condition E.
C5	3131	See 4.5.2 ($R_{\theta JC}$).
C6		Not applicable.

4.4.3.3 Group C sample selection. Samples for subgroups in group C shall be chosen at random from any inspection lot containing the intended package type and lead finish procured to the same specification which is submitted to and passes group A tests for conformance inspection. Testing of a subgroup using a single device type enclosed in the intended package type shall be considered as complying with the requirements for that subgroup.

4.4.4 Group E inspection. Group E inspection shall be performed for qualification or re-qualification only. The tests specified in table II herein shall be performed to maintain qualification.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Thermal resistance. Thermal resistance measurements shall be performed in accordance with method 3131 of MIL-STD-750. The maximum limit of $R_{\theta JC(max)}$ shall be 59°C /W. The following test conditions shall apply:

a. I_C: Collector current ..... 90 mA.

b. V_{CE}: Measurement voltage (same as V_H) ..... 10 V.

c. I_H: Collector heating current ..... 90 mA.

d. V_H: Collector-emitter heating voltage ..... 10 V.

e. t_H: Heating time ..... 1.0 s.

f. t_{MD}: Measurement delay time ..... 30 to 60 μs.

g. t_{SW}: Sampling window time..... 10 μs max.

* 4.5.3 Delta requirements. Delta requirements shall be as specified below:

Step	Inspection	MIL-STD-750		Symbol	Limit	Unit
		Method	Conditions			
1	Collector-base cutoff current 2N2218, 2N2219 2N2218A, 2N2219A 2N2218AL, 2N2218AL	3036	Bias condition D, V_{CB} = 50 V dc V_{CB} = 60 V dc V_{CB} = 60 V dc	ΔI_{CB02} (1)	100 percent of initial value or 10 nA dc, whichever is greater.	
2	Forward current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 1.0 mA dc; pulsed see 4.5.2	Δh_{FE2} (1)	25 percent change from initial reading.	

(1) Devices which exceed the group A limits for this test shall not be accepted.

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* TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1 2/						
Visual and mechanical examination 3/	2071	n = 45 devices, c = 0				
Solderability 3/ 4/	2026	n = 15 leads, c = 0				
Resistance to solvents 3/ 4/ 5/	1022	n = 15 devices, c = 0				
Temp cycling 3/ 4/	1051	Test condition C, 25 cycles. n = 22 devices, c = 0				
Hermetic seal 4/	1071	n = 22 devices, c = 0				
Fine leak						
Gross leak						
Electrical measurements 4/		Group A, subgroup 2				
Bond strength 3/ 4/	2037	Precondition T_A = +250°C at t = 24 hrs or T_A = 300°C at t = 2 hrs n = 11 wires, c = 0				
Decap internal visual (design verification) 4/	2075	n = 4 devices, c = 0				
Subgroup 2						
Collector to base cutoff current	3036	Bias condition D	I_{CBO1}		10	µA dc
2N2218, 2N2219		V_{CB} = 60 V dc				
2N2218A, 2N2219A		V_{CB} = 75 V dc				
2N2218AL, 2N2219AL		V_{CB} = 75 V dc				
Emitter to base cutoff current	3061	Bias condition D	I_{EBO1}		10	µA dc
2N2218, 2N2219		V_{EB} = 5 V dc				
2N2218A, 2N2219A		V_{EB} = 6 V dc				
2N2218AL, 2N2219AL		V_{EB} = 6 V dc				
Breakdown voltage, collector to emitter	3011	Bias condition D; I_E = 10 mA dc; pulsed (see 4.5.1)	$V_{(BR)CEO}$			
2N2218, 2N2219				30		V dc
2N2218A, 2N2219A				50		V dc
2N2218AL, 2N2219AL				50		V dc
Collector to emitter cutoff current	3041	Bias condition C;	I_{CES}		10	nA dc
2N2218, 2N2219		V_{CE} = 30 V dc				
2N2218A, 2N2219A		V_{CE} = 50 V dc				
2N2218AL, 2N2219AL		V_{CE} = 50 V dc				

See footnotes at end of table.

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* TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 4 V dc	I_{EBO2}		10	nA dc
Collector to base cutoff current	3036	Bias condition D	I_{CBO2}		10	nA dc
2N2218, 2N2219		V_{CB} = 50 V dc				
2N2218A, 2N2219A		V_{CB} = 60 V dc				
2N2218AL, 2N2219AL		V_{CB} = 60 V dc				
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 0.1 mA dc pulsed (see 4.5.1)	h_{FE1}			
2N2218				20		
2N2219				35		
2N2218A, 2N2218AL				30		
2N2219A, 2N2219AL				50		
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 1.0 mA dc pulsed (see 4.5.1)	h_{FE2}			
2N2218				25	150	
2N2219				50	325	
2N2218A, 2N2218AL				35	150	
2N2219A, 2N2219AL				75	325	
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 10 mA dc pulsed (see 4.5.1)	h_{FE3}			
2N2218				35		
2N2219				75		
2N2218A, 2N2218AL				40		
2N2219A, 2N2219AL				100		
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 150 mA dc; pulsed (see 4.5.1)	h_{FE4}			
2N2218				40	120	
2N2219				100	300	
2N2218A, 2N2218AL				40	120	
2N2219A, 2N2219AL				100	300	
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 500 mA dc; pulsed (see 4.5.1)	h_{FE5}			
2N2218				20		
2N2219				30		
2N2218A, 2N2218AL				20		
2N2219A, 2N2219AL				30		

See footnotes at end of table.

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* TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Collector-emitter saturation voltage	3071	I_C = 150 mA dc; I_B = 15 mA dc pulsed (see 4.5.1)	$V_{CE(sat)1}$			
2N2218, 2N2219					0.4	V dc
2N2218A, 2N2219A					0.3	V dc
2N2218AL, 2N2219AL					0.3	V dc
Collector-emitter saturation voltage	3071	I_C = 500 mA dc; I_B = 50 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)2}$			
2N2218, 2N2219					1.6	V dc
2N2218A, 2N2219A					1.0	V dc
2N2218AL, 2N2219AL					1.0	V dc
Base-emitter saturation voltage	3066	Test condition A; I_C = 150 mA dc; I_B = 15 mA dc; pulsed (see 4.5.1)	$V_{BE(sat)1}$			
2N2218, 2N2219				0.6	1.3	V dc
2N2218A, 2N2219A				0.6	1.2	V dc
2N2218AL, 2N2219AL				0.6	1.2	V dc
Base-emitter saturation voltage	3066	Test condition A; I_C = 500 mA dc; I_B = 50 mA dc; pulsed (see 4.5.1)	$V_{BE(sat)2}$			
2N2218, 2N2219					2.6	V dc
2N2218A, 2N2219A					2.0	V dc
2N2218AL, 2N2219AL					2.0	V dc
Subgroup 3						
High temperature operation		T_A = +150°C				
Collector to base cutoff current	3036	Bias condition D;	I_{CBO3}		10	µA dc
2N2218, 2N2219		V_{CB} = 50 V dc				
2N2218A, 2N2219A		V_{CB} = 60 V dc				
2N2218AL, 2N2219AL		V_{CB} = 60 V dc				
Low temperature operation		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 10 mA dc	h_{FE6}			
2N2218, 2N2219				15		
2N2218A, 2N2219A				35		
2N2218AL, 2N2219AL				35		

See footnotes at end of table.

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* TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 4						
Magnitude of common emitter small-signal short-circuit forward current transfer ratio	3306	V_{CE} = 20 V dc; I_C = 20 mA dc; f = 100 MHz	$\lvert h_{fe} \rvert$	2.5	12	
Small-signal short-circuit forward current transfer ratio	3206	V_{CE} = 10 V dc; I_C = 1 mA dc; f = 1 kHz	h_{fe}			
2N2218				25		
2N2219				50		
2N2218A, 2N2218AL				35		
2N2219A, 2N2219AL				75		
Open circuit output capacitance	3236	V_{CB} = 10 V dc; I_E = 0; 100 kHz ≤ f ≤ 1 MHz	C_{obo}		8	pF
Open circuit output capacitance	3240	V_{EB} = 0.5 V dc; I_C = 0; 100 kHz ≤ f ≤ 1 MHz	C_{ibo}		25	pF
Saturated turn-on time		(See figure 2)	t_{on}			
2N2218, 2N2219					40	ns
2N2218A, 2N2219A					35	ns
2N2218AL, 2N2219AL					35	ns
Saturated turn-off time		(See figure 3)	t_{off}			
2N2218, 2N2219					250	ns
2N2218A, 2N2219A					300	ns
2N2218AL, 2N2219AL					300	ns
Subgroups 5 and 6						
Not applicable						

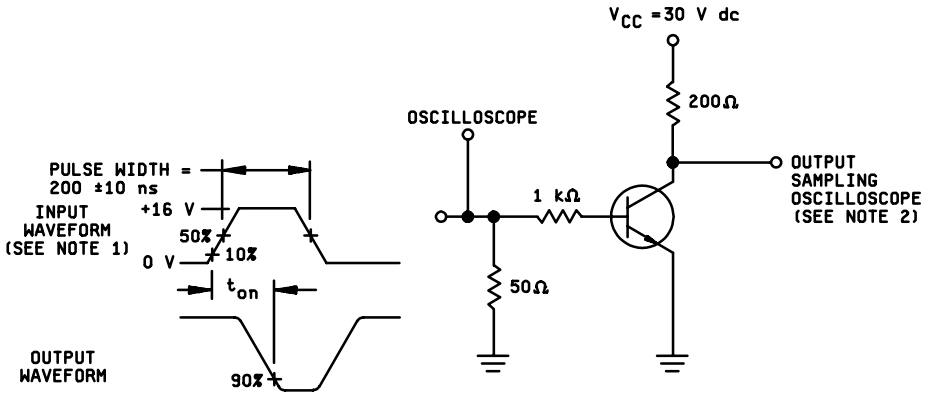
1/ For sampling plan see MIL-PRF-19500.
2/ For resubmission of failed subgroup A1, double the sample size of the failed test or sequence of tests. A failure in group A, subgroup 1 shall not require retest of the entire subgroup. Only the failed test shall be rerun upon submission.
3/ Separate samples may be used.
4/ Not required for JANS devices.
5/ Not required for laser marked devices.

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* TABLE II. Group E inspection (all quality levels) - for qualification only.

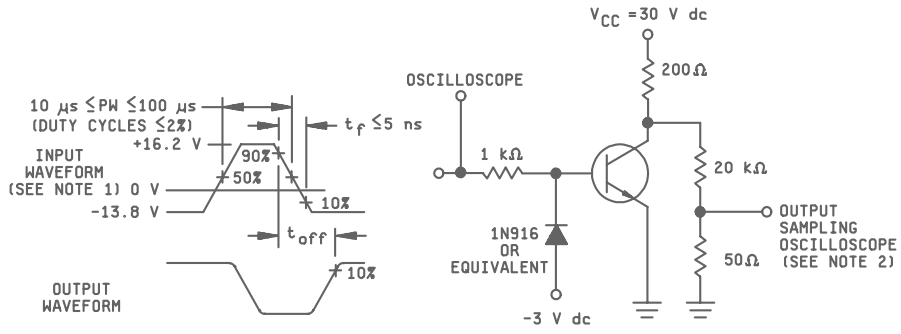
Inspection	MIL-STD-750		Qualification
	Method	Conditions	
Subgroup 1			45 devices c = 0
Temperature cycling (air to air)	1051	Test condition C, 500 cycles	
Hermetic seal	1071		
Fine leak Gross leak			
Electrical measurements		See group A, subgroup 2 and 4.5.3 herein.	
Subgroup 2			45 devices c = 0
Intermittent life	1037	V_{CB} = 10 V dc , 6,000 cycles, forced air cooling allowed on cooling cycle only.	
Electrical measurements		See group A, subgroup 2 and 4.5.3 herein.	
Subgroups 3, 4, 5, 6, and 7			
Not applicable			
Subgroup 8			45 devices c = 0
Reverse stability	1033	Condition A for devices ≥ 400 V, condition B for devices < 400 V.	

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NOTES:
1. The rise time (t_r) of the applied pulse shall be ≤ 2.0 ns, duty cycle ≤ 2 percent and the generator source impedance shall be 50 ohms.
2. Sampling oscilloscope: Z_{in} ≥ 100 K ohms, C_{in} ≤ 12 pF, rise time ≤ .2 ns.

* FIGURE 2. Saturated turn-on switching time test circuit.



NOTES:
1. The rise time (t_r) of the applied pulse shall be ≤ 2.0 ns, duty cycle ≤ 2 percent and the generator source impedance shall be 50 ohms.
2. Sampling oscilloscope: Z_{in} ≥ 100 K ohms, C_{in} ≤ 12 pF, rise time ≤ .2 ns.

* FIGURE 3. Saturated turn-off switching time test circuit.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2).

c. Packaging requirements (see 5.1).

d. Lead finish (see 3.4.1).

* 6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers' List (QML) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43216-5000.

* 6.4 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 11
NASA - NA
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Review activities:
Army - AR, MI, SM
Navy - AS, MC, SH
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Preparing activity:
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-19500/251K	2. DOCUMENT DATE 3 June 2002

3. **DOCUMENT TITLE** SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, SWITCHING TYPES 2N2218, 2N2218A, 2N2218AL, 2N2219, 2N2219A, AND 2N2219AL JAN, JANTX, JANTXV, AND JANS

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) COMMERCIAL DSN FAX EMAIL	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. Point of Contact Alan Barone	b. TELEPHONE Commercial: 614-692-0510 DSN: 850-0510 FAX: 614-692-6939 EMAIL: alan.barone@dscc.dla.mil
c. ADDRESS Defense Supply Center Columbus ATTN: DSCC-VAC P.O. Box 3990 Columbus, OH 43216-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman, Suite 2533 Fort Belvoir, VA 22060-6221 Telephone (703) 767-6888 DSN 427-6888

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